



Complex Systems

2021/2022

Exam N 22 of June 2022 Time: 2h00

Name:

Number:

This is a **closed books** exam. Any attempt at cheating implies immediate disapproval in the course and the activation of the procedures defined in the disciplinary regulation document of the UC.

Question #	Points	Your score
1	18	
2	12	
3	25	
4	25	
5	20	
	100	

Grader:

Question 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. If applicable, you may use a concrete example to **help you** to explain the concept.

a) Complex System.

Your answer:

b) Segregation Model.

Your answer:

c) Diameter (in a graph).

Your answer:

Question 2 - **Comment** 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) The dynamical system $\frac{dx}{dt} = f(x)$ cannot have an **oscillatory** behavior whatever form $f(x)$ may take. .

Your answer:

Your answer:

- b) In agent-based modeling a Moore's neighborhood is preferable to a Von Neumann neighborhood.

Your answer:

Your answer:

Question 3 - Simulate 25 points

Suppose that you have just finished preparing a soup and place it on the table. The soup is very hot and ~~so~~ you have to wait until it cools down. As you are a dynamical systems scientist you decided to define a **model of the cooling process** using differential equations. **What was your solution?** Describe, and justify, the **assumptions** you made in your model. Draw the corresponding **phase space** and identify the **fixed-points** and their nature, if any.

Your answer:

Your answer:

Question 4 - Explain 25 points

During the course we discussed three major models to dynamical systems: mathematical models (difference equations, differential equations), agent-based models, and network models. In the case of modeling a virus infectious disease, what are the advantages (pros) and disadvantages (cons) of each approach? These characteristics can be generalized to other types of modeling problems?

Your answer:

Your answer:

Question 5 - Model 20 points

In class we briefly mentioned a model, based on a system of differential equations, about a **love affair**, called Romeo and Juliet. The model was intended to describe the dynamics of the love between two persons. It has two variables and four parameters. The concrete values of these parameters make possible the definition of different types of love relationship. Show the model and define values for the parameters so to describe a **cautious lover**, i.e., one that is encouraged by the other positive feelings but retreats from his own feelings.

Your answer:

Your answer:

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